

Advancing Frontiers of Energy Forecasting with Novel Methods



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DISCLAIMER

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ABSTRACT

Regional transmission organizations must continuously balance electricity supply and demand, but day-ahead forecasts are often least reliable during extreme weather. This project tested whether Adaptive Conditional Neural Processes (AdaCNP) improve extreme-load forecasting over a standard Conditional Neural Process baseline on ERCOT data. Evaluation used a leave-one-event-out protocol across 17 extreme cold events from 2002–2026, with seven-day exclusion buffers, a 9:00 a.m. CT issuance cutoff, and documented loadshed hours removed from the primary metric.

The pre-registered comparison found an AdaCNP advantage in Gaussian negative log likelihood, with a mean paired difference of 0.459, a 95 percent confidence interval of $[0.108, 0.811]$, and $p = 0.014$. However, the effect was slightly below the study's minimum detectable effect, appeared only in the pre-registered configuration, and could not establish an effect as small as the one reported in the source study.

The most important operational result was poor calibration during extreme events. A nominal 90 percent interval covered 96 percent of normal-period hours but only 63 percent of held-out extreme-event hours. Both models systematically underforecast demand while reporting high confidence, and coverage worsened as event severity increased. AdaCNP also placed less weight on cold historical analogs than uniform weighting did, so its improvement was not explained by the proposed mechanism. These results suggest that improving extreme-event calibration is more important for reserve planning than further architecture comparison.

I. INTRODUCTION

Electricity cannot be stored economically at grid scale, so a balancing authority must match generation to demand continuously. That obligation forces a decision under uncertainty every day: commit generating resources today for demand that will materialize tomorrow. Undercommit and reserves fall short; overcommit and consumers pay for capacity that idles. Regional transmission organizations and independent system operators¹ are permitted to commit resources early, before the day-ahead market, to hedge extreme weather, but doing so runs against their mandate to dispatch at least cost. The value of that hedge depends entirely on whether the forecast's stated uncertainty can be trusted. Probabilistic load forecasting addresses this problem by describing a range of possible outcomes rather than providing only a single point estimate.¹ Large-scale forecasting competitions have similarly demonstrated the importance of evaluating both forecast accuracy and uncertainty.²

Extreme weather is where this is hardest and where it matters most. The February 2021 winter storm drove ERCOT to shed firm load. Such events are also rare by construction, which creates a data problem: a model may have tens of thousands of ordinary days to learn from and only a handful of the days that actually determine reserve adequacy. Global probabilistic forecasting methods such as DeepAR can share statistical information across related time series, but rare regime changes and extreme events remain difficult to represent reliably.³ Worse, the relationship between weather and demand is not stable across regimes, so ordinary days can actively mislead.

This project addressed that problem by testing a published method on new data. Hu *et al.*⁴ propose Adaptive Conditional Neural Processes, which extend Conditional Neural Processes by weighting historical context examples according to their similarity to the current forecast target rather than averaging them uniformly. Conditional Neural Processes were developed as flexible probabilistic models that learn distributions over functions from context observations.⁵ Attentive Neural Processes later incorporated attention mechanisms to allow the model to place different amounts of emphasis on individual context observations.⁶ Hu *et al.* report improved extreme-load forecasting on the PJM and ISO-NE systems. The goal of this project was to determine whether that reported advantage reproduces on ERCOT data and to establish what such a test can and cannot demonstrate given the number of extreme events ERCOT's history supplies.

Project boundaries were fixed at the outset. The comparison is AdaCNP against standard CNP only, on a shared encoder-decoder backbone, with the aggregation mechanism as the single intended difference; no other architectures were evaluated. Evaluation is restricted to held-out extreme cold events under a leave-one-event-out protocol. All work is retrospective on historical data; nothing was deployed operationally.

II. METHODOLOGY

A. Data and event definition

The target series is hourly ERCOT system load from 2002 to 2026, adopted by content hash rather than by filename so that provenance is exact. Extreme events come from a frozen inventory derived from a regional temperature index, with severity characterized by cold margin in degrees Celsius. Seventeen events are load-eligible, meaning they fall within the span of the load record.

¹Regional transmission organizations and independent system operators coordinate wholesale electricity markets and help maintain the real-time balance between electricity supply and demand within their respective regions.

This count is derived from the artifacts at run time rather than hard-coded, so the protocol adapts if the inventory changes.

An episode targets one calendar day and predicts its 24-hour load vector. Inputs are calendar terms, the 24 most recent complete hourly load observations, the 24 most recent temperature observations, the 24-hour rolling mean temperature at the cutoff, and nonlinear heating and cooling terms: 57 features in total. Every feature is restricted to information complete at or before the day-ahead issuance instant, fixed at 9:00 a.m. America/Chicago on the day preceding the target day.

B. Leakage control and censored observations

Three protections were built and verified by automated test rather than by inspection. Outer partitions are leave-one-event-out with a seven-day buffer on each side of the held-out event, excluded from both training and context retrieval. Both model arms consume a single persisted context-index file, reread from disk by each arm, so their context sets are byte-identical by construction. Normalization statistics are fitted on each fold's training partition only.

Load measured during directed load shed is a lower bound on the demand that would otherwise have materialized, not an observation of it. Across the 1,271 event-hours in the record, 80 hours are documented as shed, zero hours are documented as not shed, and the remaining 1,191 carry no primary-source determination either way. The adopted treatment excludes documented shed hours from the primary metric, retains undetermined hours while flagging them, and reports an all-hours served-load likelihood as a secondary diagnostic on a different estimand. Resolving the undetermined hours in either direction would assert a finding the record does not support.

C. Statistical design and pre-registration

The primary metric is Gaussian negative log likelihood on held-out event-period load.² Because normalization is fitted per fold, likelihood values are not comparable across folds; the only admissible quantity is the within-event paired difference between arms, which share a fold's normalizer and context file. The unit of analysis is therefore one paired difference per event with the three frozen seeds averaged, giving a sample size of 17.

The analysis was pre-registered. The primary endpoint, the configuration to be tested, the statistical test, the significance level, the robustness checks, and the decision rule, including what a null result would be permitted to claim, were committed to version control before any confirmatory run executed. The analysis code was committed at the same time and validated against earlier exploratory data, so the reported number was produced by code whose commit provably predates the data it analyzed. The tested configuration was chosen for fidelity to Hu *et al.*, specifically by using temperature features and randomly sampled context sets that matched their inputs and sampling procedure. It was explicitly not chosen for the size of the effect observed in exploratory work.

A power analysis preceded the confirmatory runs, using variance measured from exploratory data rather than assumed. It established the minimum detectable effect at 80 percent power, reported alongside every result so that a null can be distinguished from an absence of power.

The full sweep comprised 408 runs: 17 folds by 2 arms by 2 feature sets by 2 context conditions

²Negative log likelihood evaluates both the location and the uncertainty of a probabilistic forecast. Lower values indicate that the model assigned greater probability to the observed outcome.

by 3 seeds. The sweep was executed on CPU with zero failures. Training length was not fixed by hand: each run trained to a ceiling, and the reported parameters were those achieving the best inner-validation likelihood, with inner validation drawn from the fold's own training partition. Selecting on the held-out event would have constituted leakage.

III. RESULTS AND DISCUSSION

A. Calibration fails during extreme events, in the dangerous direction

The most operationally consequential result is not the comparison between architectures. It is that both models' stated uncertainty is untrustworthy precisely in the regime the project set out to study.

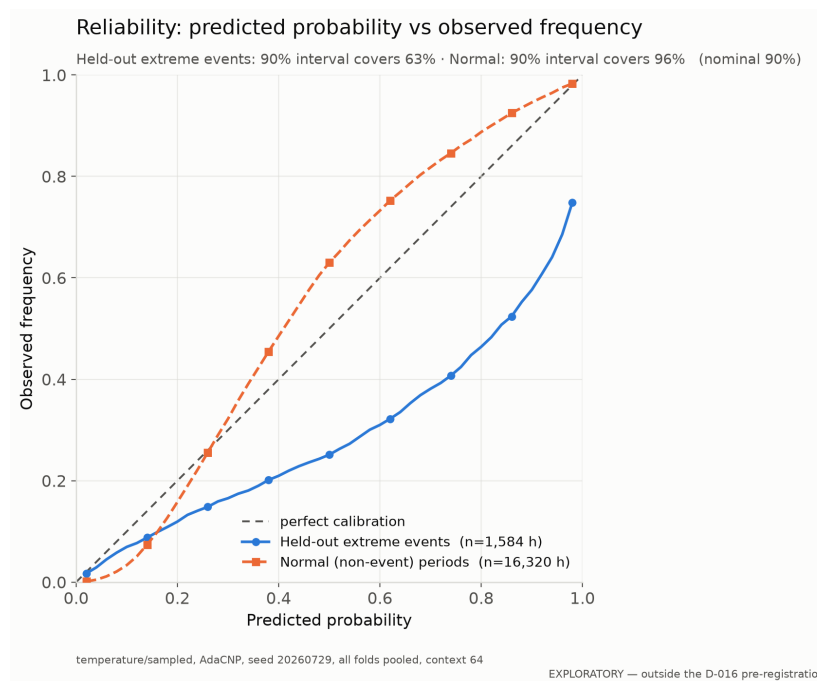


FIG. 1. Reliability of the predictive distribution, by regime. For each nominal probability, the model's corresponding quantile is computed and the observed frequency of outcomes at or below it is plotted; a perfectly calibrated forecast follows the diagonal. Both curves come from the same trained models, evaluated on held-out extreme-event days and on non-event days drawn from the same folds' inner-validation blocks. The 90 percent central interval contains the observed load for 96 percent of normal-period hours but only 63 percent of extreme-event hours. **Exploratory: outside the pre-registered analysis.**

Figure 1 shows the failure. In normal conditions, the forecast is honest and slightly conservative, its intervals marginally too wide, which is the safe direction in which to err. During held-out extreme events, the same models' 90 percent interval covers 63 percent of hours. The extreme-event curve also lies below the diagonal across its entire range, meaning fewer outcomes fall below each predicted quantile than the model expects: the models systematically underforecast demand during extreme cold while reporting high confidence. This is the error mode that would lead an operator to skip an expensive early commitment.

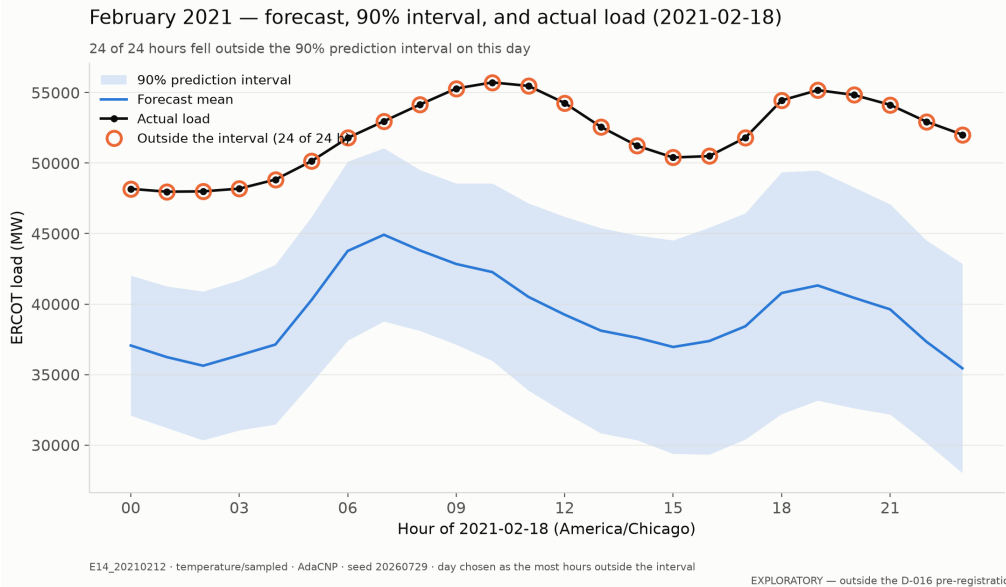


FIG. 2. A single day of the February 2021 event. The day shown was selected by a stated rule: it is the day of that event with the most hours falling outside the 90 percent interval. It is therefore the worst case, not a typical day. All 24 hours fell outside the interval, with observed load running roughly 10 GW above the forecast mean. The day carries no documented load-shed hours, so the gap reflects forecast error rather than curtailed demand. **Exploratory.**

Figure 2 makes the magnitude concrete for one day of the February 2021 event. Figure 3 shows that the deficit is not uniform: coverage declines as events become more severe, at approximately -0.037 per degree Celsius of cold margin for CNP ($r = -0.55$) and -0.031 for AdaCNP ($r = -0.51$). Only the mildest events approach nominal coverage. For practice, the implication is that calibration is worst exactly where reserve margins are thinnest.

B. The pre-registered comparison

The pre-registered test detected an advantage for AdaCNP on held-out ERCOT extreme events; Table I reports it.

TABLE. I. Pre-registered primary result. Positive differences favor AdaCNP.

Quantity	Value
Events (n)	17
Seeds per event	3
Mean paired difference	+0.459
95% confidence interval	[+0.108, +0.811]
Two-sided paired t -test	$p = 0.014$
Wilcoxon signed-rank test (robustness)	$p = 0.020$
Minimum detectable effect	0.495

Predeclared robustness checks are consistent. The Wilcoxon signed-rank test agrees with the

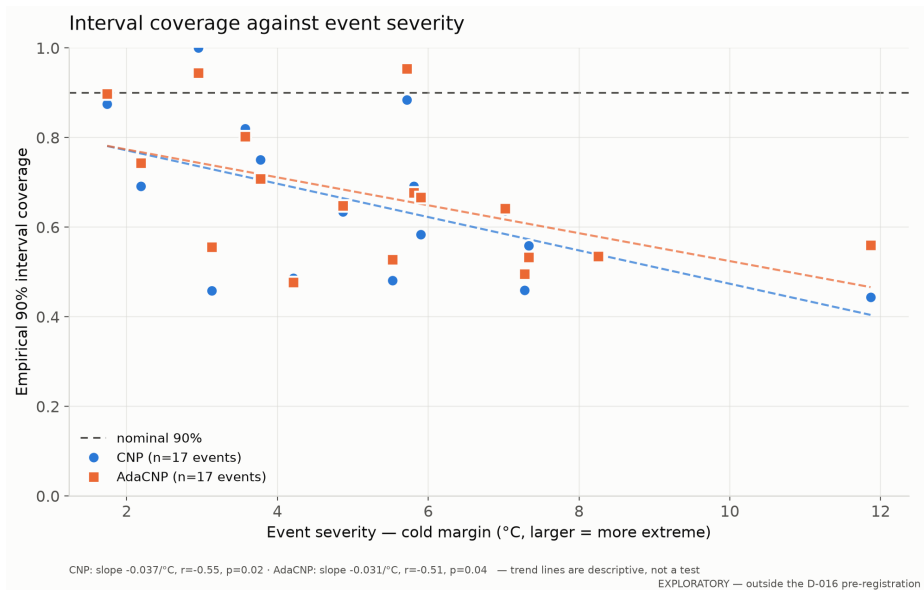


FIG. 3. Empirical 90 percent interval coverage against event severity across all 17 leave-one-event-out folds and both arms. Trend lines are descriptive and form no part of the pre-registered analysis. **Exploratory.**

t -test, so the pre-registered inconclusive branch was not triggered. Excluding the one fold reduced to a single held-out day gives a mean of $+0.486$ ($p = 0.014$). A leave-one-fold-out jackknife spans $[+0.343, +0.501]$, with the sign stable across all 17 refits, so no single event drives the finding, and 11 of 17 events favor AdaCNP. Secondary metrics point the same way: continuous ranked probability score³ 0.469 against 0.486, root mean squared error 0.756 against 0.773, and calibration error 0.247 against 0.272. The frozen protocol reserves adjudication to the primary metric, so these are reported for interpretation only.

Three qualifications belong with that result rather than in a footnote.

First, the observed effect of $+0.459$ lies just below the minimum detectable effect of 0.495. The study therefore had roughly 77 percent power at the effect size it actually found, short of the 80 percent target. The result satisfies the pre-registered decision rule, but it sits near the edge of what this design can resolve and warrants independent replication rather than treatment as settled.

Second, the advantage appears only in the pre-registered configuration. Of the four configurations run, the pre-registered one gave $+0.459$; the same feature set with nearest-neighbor context retrieval gave -0.240 , favoring the baseline. Had the configuration been chosen after inspecting results, a range of conclusions could have been supported. The pre-registration is what makes the reported claim admissible, and that dependence is stated deliberately.

Third, this is not a measurement of the effect size Hu *et al.* report. Their AdaCNP-over-CNP likelihood margin corresponds to roughly 0.02 to 0.12 in the units used here, an order of magnitude below this design's minimum detectable effect. The power analysis puts this study's power to detect a margin that size at 5 to 9 percent and estimates that several hundred events would be required; ERCOT's history supplies 17. A null would consequently have been uninformative about

³The continuous ranked probability score compares the complete predictive distribution with the observed outcome. Lower values indicate a better probabilistic forecast.

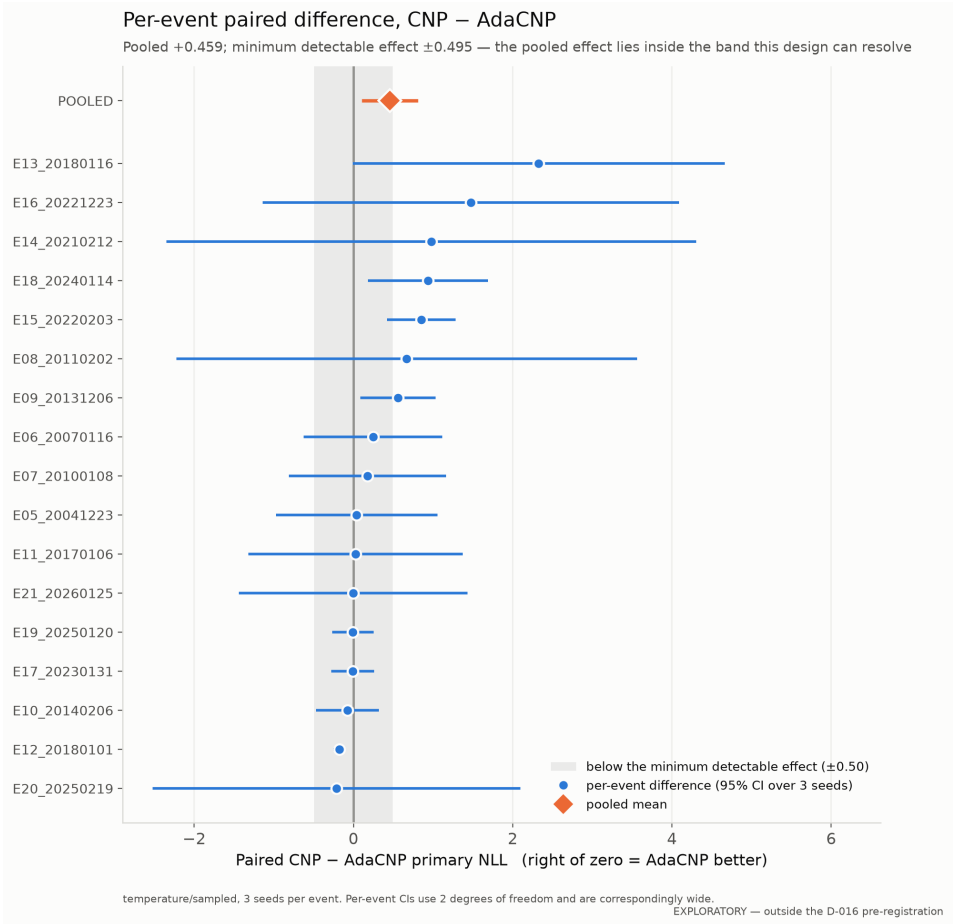


FIG. 4. Per-event paired differences with 95 percent confidence intervals, the pooled mean, and the shaded band below the minimum detectable effect. Per-event intervals use two degrees of freedom and are correspondingly wide; they indicate shape, not per-event significance. **Exploratory presentation of the pre-registered quantity; the confirmatory result is Table I.**

the source finding, which is precisely why the minimum detectable effect is reported beside the estimate.

C. The mechanism does not behave as described

AdaCNP's proposed advantage comes from concentrating attention on relevant historical analogs, and during extreme cold that means similarly cold days. This is directly measurable once temperature features are present. Under uniform weighting, CNP places 0.105 of its context weight on cold days, which is simply the cold fraction of its context set. AdaCNP places 0.079: it deemphasizes cold analogs, and its weighting is only mildly nonuniform, using 57.5 effective context points out of 64.

AdaCNP therefore scores better on held-out extreme events, but not by doing what its authors describe. Whatever produces the improvement observed here, it is not concentration on cold analog days. This is an open question and, for future work, a more interesting one than the score difference itself.

D. Limitations

The event count is fixed at 17 by ERCOT's history and by the extreme-weather framing; the power analysis identifies it as the binding constraint. Of the 1,584 held-out hours scored per seed, 1,139 carry undetermined load-shed status, so the primary metric carries an unknown amount of censoring contamination that the record cannot resolve. No day-ahead forecast weather product with historical issuance timestamps was available, so all temperature features are past-observed; Hu *et al.* additionally use next-day forecast temperature, and this remains the largest divergence from their protocol. Three seeds were used, initialization noise dominates the measurement, and additional seeds would be the cheapest available improvement to precision. Finally, only the primary endpoint is confirmatory; the calibration results and the mechanism measurement are exploratory and labeled as such throughout.

IV. CONCLUSIONS

The pre-registered comparison detected an advantage for AdaCNP over a standard CNP baseline on held-out ERCOT extreme cold events: +0.459 in Gaussian negative log likelihood, a 95 percent confidence interval of $[+0.108, +0.811]$, $p = 0.014$, and $n = 17$, with robustness checks consistent and no single event driving the result. The advantage is real by criteria fixed in advance, but it sits near the resolution limit of the design, it depends on the pre-registered configuration, and it is not a measurement of the effect size reported in the source work.

The more consequential finding for grid operations is that both models are badly and directionally overconfident during extreme events. A nominal 90 percent interval that contains the outcome 96 percent of the time in normal conditions contains it 63 percent of the time during extreme cold, with the deficit widening as events worsen. For a balancing authority weighing early generation commitment, a forecast that is uncertain and says so is manageable; a forecast that is wrong and confident is not. On this evidence, improving calibration in the extreme regime would repay more than further architecture comparison.

A. Summary of appointment activities

Over the appointment, I built a complete and reproducible experimental pipeline for probabilistic extreme-load forecasting on ERCOT data: verified data import with content-hash provenance, leakage-controlled leave-one-event-out partitioning with issuance-time enforcement, an adopted treatment for censored load observations, a CPU-based training and evaluation harness for both model arms, a pre-registered statistical analysis, and a power analysis quantifying the design's resolution. Activities included a literature review that identified the source method, reconstruction and hash verification of the project's inherited data corpus, a documented audit comparing this implementation against the source publication, and execution of a 408-run confirmatory sweep. Decisions of record are captured in a versioned decision log, and every reported figure is regenerated from committed artifacts by script rather than transcribed.

B. Related activities and publications

Related activities included regular technical discussions with my mentor and project team, review of relevant scientific and technical literature, preparation and delivery of an end-of-summer presentation, development of the final research report, and participation in SULI and NETL learning and professional-development activities. These experiences strengthened my skills in technical communication, research planning, scientific writing, reproducibility, data analysis, and presentation of research findings to both technical and general audiences. No publications resulted from this appointment.

ACKNOWLEDGMENTS

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